Interactive System for Origami Creation

Takashi Terashima, Hiroshi Shimanuki, Jien Kato, and Toyohide Watanabe

Graduate School of Information Science, Nagoya University
Furo-cho, Chikusa-ku, Nagoya 464-8601, Japan
{takashi, simanuki}@watanabe.ss.is.nagoya-u.ac.jp,
{jien, watanabe}@is.nagoya-u.ac.jp

Abstract. This paper proposes a new system which supports origami creators who have no special knowledge about origami creation to create their unique works easily in 3-D virtual space. Moreover, 2-D diagrams or 3-D animation are automatically made for describing the folding processes so that people can re-build these works. Users can decide folding operations and create works by an interactive interface. For easy creation, two methods are proposed. One is a method for representing overlapping-faces of 3-D virtual origami in order to support users' recognition of origami's conformation. As a result, users can input information about folding operations easily and correctly. The other one is a method for deriving halfway folding processes according to users' intents. Even if users have rough images about shapes of origami works, they may not be able to start creating an origami model as their imagination. Namely, the system shows folding processes from square to basic forms until they can start do it by themselves. We expect that the common people will create and publish their unique works and more people will enjoy origami.

Keywords: Origami, Interactive Interface, Computer Graphics, 3-D Virtual Model, Origami Base.

1 Introduction

Origami, one of the Japanese traditional cultures, is perceived worldwide as the art of paper folding which has abundant potential. Making origami assists not only the enhancement of concentration and creativity but also rehabilitation exercise, antiaging effects, and so on. Traditionally, people play origami based on drill books (text books) or materials on web pages [1] in which the folding processes consist of simple folding operations are illustrated by diagrams. Recently, a system which recognizes folding operations from origami drill books and displays 3-D animation of folding processes were proposed [2] [3]. On the other hand, these drill books or materials are made and exhibited by limited persons who have special knowledge about origami creation. It is difficult for the people who have no special knowledge about origami creation to create their unique works and to describe the folding processes by diagrams so that people can re-build them (i.e. to publish works). The main reason of this is botheration of using tangible papers thorough trial and error processes. Another reason is trouble of making drill books or other instructional materials. For these reasons, few people create new

W. Liu and J. Lladós (Eds.): GREC 2005, LNCS 3926, pp. 182–194, 2006.
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origami works and it is not often that innovative works are made in public. Therefore, an environment that facilitates creative activities is required.

This paper proposes an Interactive System for Origami Creation. This system supports origami creators including the people who have no special knowledge about origami creation. Users can transform virtual origami by operating this system interactively. Using the system, they are able to create their unique works easily and comfortably. Moreover, 2-D diagrams or 3-D animation which describe the folding processes can be automatically made for publishing. We expect that the common people will create and publish their unique works and more people will enjoy origami. As related work, a system that represents dialogical operations of origami in 3-D space has been introduced [4]. However, origami creation is not considered by using the system.

In order to let users input their intended operations without any mistakes or difficulty, we consider a user interface and some functions which ease users' operations and help their recognition about shape of 3-D virtual origami. Hereafter, we first show the framework and user interface of this system in section 2. Then, two methods for improving the usability of our system are proposed in section 3 and section 4. One of the methods is for representing virtual origami. The other is for deriving halfway folding processes. Finally, we show the conclusions and future prospects in section 5.

2 Framework and Interface

In this section, we show the framework of our proposed system and outline the system. Then, we show user interface and consider how to improve the usability of the system.

2.1 Framework

Figure 1 and Figure 2 show the basic framework of the system and interaction between the system and a user, respectively. The system first receives positional information of a fold line determined by user's input. Then, the feasible folding operations are constructed based on the crease information, which is superficial and incomplete. They are obtained by maintaining consistency of crease patterns under some geometrical constrains [5]. All the feasible candidates are simulated against an internal model of origami. As a consequence, several different origami states are obtained from each candidate. Subsequently, the system presents resultant models corresponding to those candidate operations. Finally, the user selects his/hers desired operation. In this way, this interaction enables users to input folding operations easily. Namely, users can transform virtual origami variously by the basic mouse action. By the repetition of the interaction, the system can understand a sequence of folding operations required to create an origami work, and represent them in the form of 3-D animation or a sequence of 2-D diagrams.

2.2 User interface

Figure 3 shows user interface of proposed system. A state of origami at some step is displayed on the left of the window, while the states which are simulated according to candidate operations (see the previous section) are displayed on the right of the window.

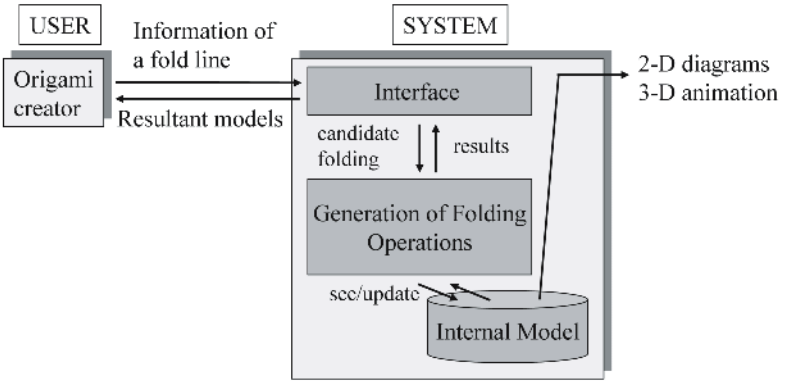
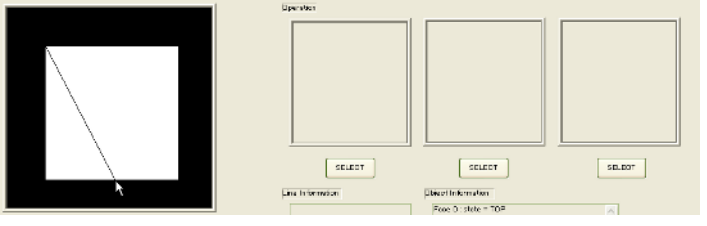
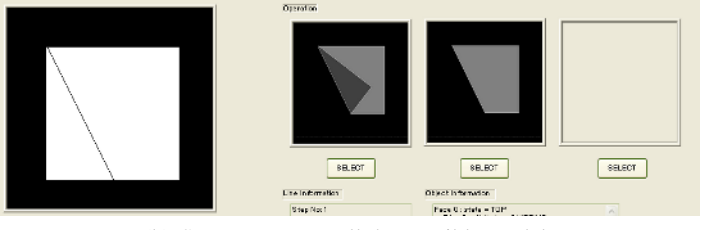


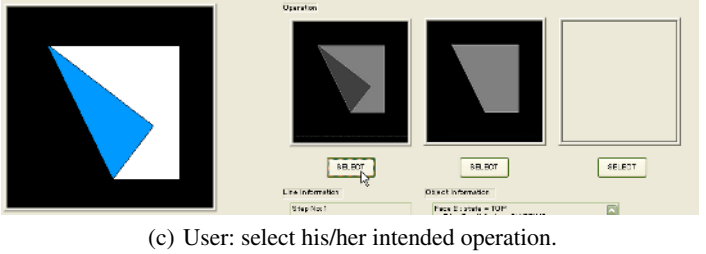
Fig. 1. Framework of proposed system



(a) User: input a fold line.



(b) System: present all the possible models.



(c) User: select his/her intended operation.

Fig. 2. Interaction for folding operation decision

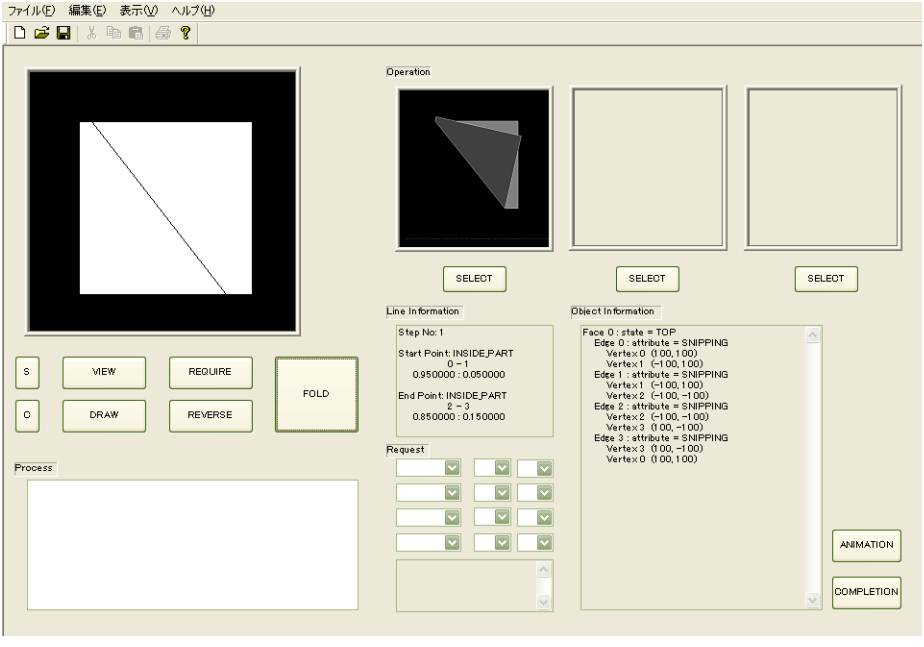


Fig. 3. User interface

The left graphic has two modes, view mode and draw mode. In view mode, users can see the origami model from all viewpoints and can not input anything. On the other hand, in draw mode, users can draw a fold line from fixed viewpoint. By the idea having two modes, users can understand the shapes of origami model and can draw a fold line correctly.

Generally, there are several considerations to improve the usability of the system. In order to design an ideal user interface for easy-to-use system, we discuss three elements: intended users, cognitive load, and operational error.

Intended Users. People often feel that to create origami works with real paper is too much trouble, for example, paper crumples up through a trial and error process. Moreover, it is difficult to remember the folding processes for the created works, and also difficult to describe the folding processes in a sequence of diagrams for publication. From these backgrounds, the aim of intended users of the system is to create and to publish their unique origami works comfortably and easily by using the system. Additionally, we assume that intended users do not have special knowledge about origami creation (such as design knowledge based on a crease pattern) and they have rough images about shapes of origami works (such as a four-legged mammal).

Cognitive Load. There are various kinds of folding operations. Since users have to give the desired folding operations correctly, an environment which enables users to understand the configuration of origami intuitively and to input folding operations by simple actions is required.

As mentioned previously, our proposed system enables users to input various kinds of folding operations through the interaction between the system and users. At this time, users' required action is only to input folding operations through the basic mouse action. Furthermore, users can see an origami model from all viewpoints in view mode.

Operational Error. There is a possibility that users draw a fold line at the wrong (undesired) position. We should consider preventive measures and countermeasures against this operational error.

As a preventive measure, an environment which enables users to understand the configuration of origami intuitively (mentioned in section 2.2) is required. Furthermore, as a countermeasure, the system has an undo/redo function which allows users to undo their inputs from any step in case of operational error.

2.3 Required Methods

From these discussions, we must propose following two methods. One is a method for representing 3-D virtual origami, discussed in section 2.2 and 2.2. Generally, an origami model is constructed by planar polygons corresponding to faces of origami. Therefore, when an origami model is displayed, multiple faces on the same plane (called overlapping-faces) probably seem to be one face. This incorrect perception occasionally obstructs users' inputs. Consequently, we must propose a method for representing overlapping-faces of 3-D virtual origami in order to support users' recognition of origami's conformation in both view mode and draw mode. As a result, users can input information about folding operations easily and correctly.

The other one is a method for deriving halfway folding processes. Even if users have rough images about shapes of origami works as mentioned in section 2.2, they may not be able to start creating an origami model as their imagination or may not be able to continue at a step, especially when they do not have special knowledge about origami creation. In order to deal with such case, we must propose a method for deriving halfway folding processes according to users' intents. Namely, the system shows folding processes to users until they can start do it by themselves.

In this paper, we describe these methods in detail. The former method is proposed in section 3, while the latter method is proposed in section 4.

3 Method for Representing Origami

This section specifically describes our method for representing overlapping-faces of 3-D virtual origami for the user interface.

3.1 Our Approach

In order to represent virtual origami 3-dimensionally, we consider the extended (i.e. ideal) representation as the reconfiguration of a 3-D origami model. Specifically, overlapping-faces are moved apart slightly by rotating polygons along a rotation axis determined from figurations and relationships of faces. Because of the reconfiguration in 3-D space before 3-D rendering, this method has the advantage that an origami model

can be seen from all viewpoints without any renewed reconfigurations if once it is reconfigured. Namely, the reconfiguration depends not on users' viewpoints, but on the origami model.

The elementary transformation is a movement (i.e. rotation) targeted at two faces which are adjoining each other. Order and portions of movement are based on figurations and relationships of overlapping-faces. We discuss which faces should be moved, which portions of the faces should be rotated, and what order they should be rotated in.

3.2 Free-Portion

We define a "free-portion" (part of a face) as the portion that is not restrained by the adjoining face and can move freely. Such free-portions should be moved (i.e. rotated). In order to find out a free-portion, firstly, we define a "free-edge" as follows.

Free-Edge. Given two faces (F_1 and F_2) that are on the same plane and are adjoining each other, an edge E of F_2 is a "free-edge" to F_1 if following conditions are all satisfied:

- E is not an edge of F_1, but an edge of F_2.
- E and F_2 are not covered by other faces.

In order to determine whether the latter condition is satisfied, cross-sections of origami are generated by cutting origami perpendicular to E. Figure 4 gives examples of free-edge and not-free-edge. At State C, both sides of $F2$ are covered by other faces, and these faces are joined on the same side of E. Namely, E and F_2 are not covered by other faces, and E is a not-free-edge. This definition is used to determine free-portion as follows.

Free-Portion. A free-portion of a face F_2 to the adjoining face $F1$ is determined by following steps.

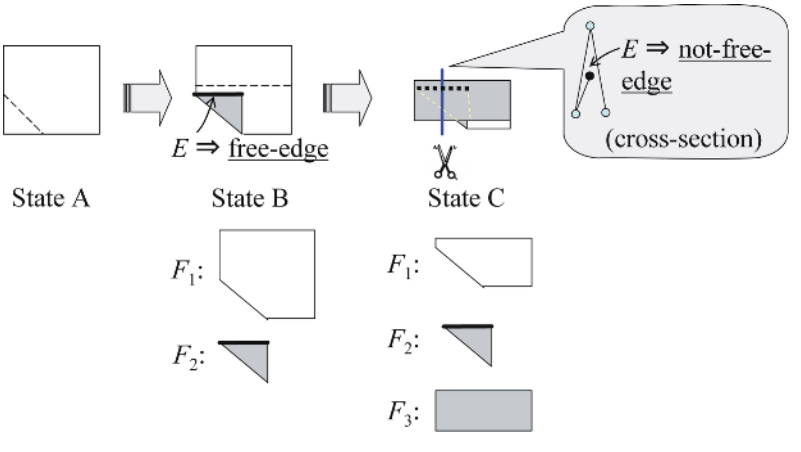


Fig. 4. Examples of free-edge and not-free-edge

1. Determine whether each edge of F_2 is a free-edge to F_1.
2. Draw a line L that connects two points between free-edge and not-free-edge.
3. Define the polygonal area enclosed by the free-edges and L as a free-portion.

This line L becomes the rotation axis when the free-portion is rotated. Figure 5 show examples of determining free-portion. In the case of F_1, the free-portion is the triangular shape (i.e. half of F_1). On the other hand, in the case of F_2, the free-portion is the whole face F_2. More specifically, F_2 is unrestrained in moving by F_3. In addition to these examples, there are cases where no free-portions of some faces exist since polygonal area can not be formed in step 3.

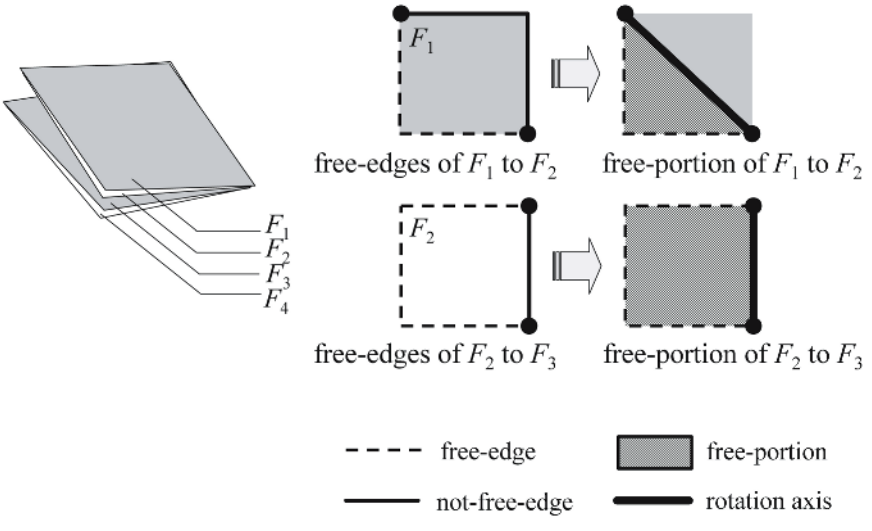


Fig. 5. Examples of determining free-portion

3.3 Grouping of Faces

In Figure 5, the free-portion of F_3 to F_4 is the triangular shape like that of F_1 to F_2. If the free-portion of F_3 is rotated before the rotation of F_2 (whole area is the free-portion), the free-portion of F_3 collides against F_2 and the reference plane of the rotation of F_2 gets fuzzy.

To solve this problem, we propose a method that groups overlapping-faces based on dependency relation about their movements. Namely, if a face can move independently of another face, the two faces are classified into different groups. Otherwise, they are classified into the same group. This grouping of faces determines the order of face's movements. The procedure for grouping overlapping-faces is described as follows.

Procedure for Grouping.

1. Make the order list of overlapping-faces.
2. Determine free-portion of each face to the adjoining face behind it (beginning at the bottom).

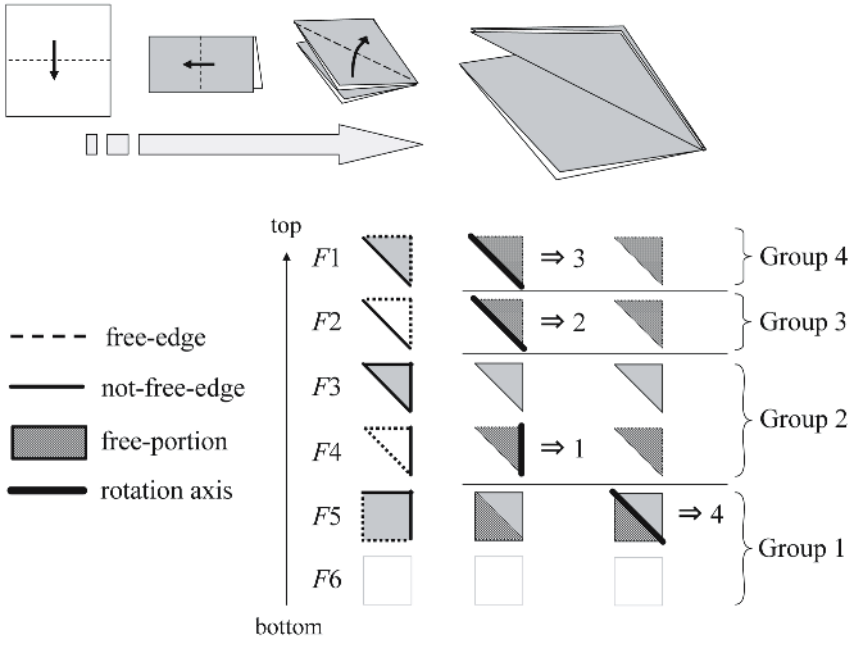


Fig. 6. Examples of grouping faces

3. Let the faces that whole area is the free-portion be chief faces of their groups. Let the undermost face also be chief face.
4. Crassify each not-chief face into the group the nearest behind chief face belongs to.

This grouping solves the problem described above. More specifically, no faces collide against other faces by moving all faces which belong to the same group before the rotation of each free-portion. Each rotation angle can be decided in consideration of angular difference between anteroposterior groups.

3.4 Representation Algorithm

Our proposed method for representing 3-D virtual origami is summarized as follows.

Representation Algorithm.

1. Make the order list of overlapping-faces.
2. Determine free-portion of each face to the adjoining face behind it (beginning at the bottom).
3. Determine chief faces and classify other faces with appropriate groups.
4. Rotate set of faces in each group collectively along the chief's axis (i.e. chief face of the group and faces which belong to the group). Rotation angle is constant.
5. Rotate free-portions of overlapping-faces in sequence along respective axes.

Figure 6 shows example of representing 3-D origami based on this algorithm. In this case, four chief faces and four groups are formed. Subsequently, sets of faces in

group 2, group 3, and group 4 are rotated along their chiefs' axes. Finally, the free-portion of F_5, only not-chief face which can move, is rotated along own axis.

4 Method for Deriving Halfway Folding Processes

If users can not start or continue folding virtual origami, the system should show folding processes to users until they can do it. In this section, we propose a method for deriving halfway folding processes according to users' intents.

4.1 Our Approach

It is sure that users who have rough images about shapes of origami works have the most difficulty in folding virtual origami from square to some step. For example, when a user wants to create a four-legged mammal (such as a dog), can he/she specify the first operation of the folding process? Moreover, can he/she know how to fold to make six corners which will become four legs, a head, and a tail eventually? The answers to these questions are probably "No" if the user does not have special knowledge about origami creation.

Noting this, we propose a method for deriving folding processes from square to some step so that users can start creating origami. In the case of above example, the system should derive and show the folding process until six corners are composed. After that, in order to create a dog, the user will be able to fold origami to determine corners' positioning, balance, and so on.

4.2 Origami Base

We use the idea of an origami base [6, 7] in our deriving method. An origami base is a specific form at the intermediate stage of folding origami from square (initial state) to the specific work. The base has about the same number of corners as the corresponding work. Figure 7 shows an example of an origami base. Crane base has five corners corresponding to five parts of the work: crane's head, tail, body, and two wings. Furthermore, various works can be created from one common origami base. In Figure 7, works which have about five parts can be derived from crane base. There are about twenty origami bases, and most origami works are derived from one of them.

Each origami base has several long and short corners. Moreover, corners can be grouped based on their constructional symmetry. For example, in the case of crane base (see Figure 7), there are four long corners and one short corner. These four long corners are divided into two groups: the group of two upward corners (called group A) and that of two downward corners (called group B). As above, corners of an origami base have two attributes, length and symmetry.

4.3 Supporting Origami Creation Based on Origami Base

As mentioned previously, parts of origami works are closely associated with corners of origami base. Therefore, when users have intents about parts of origami works, the

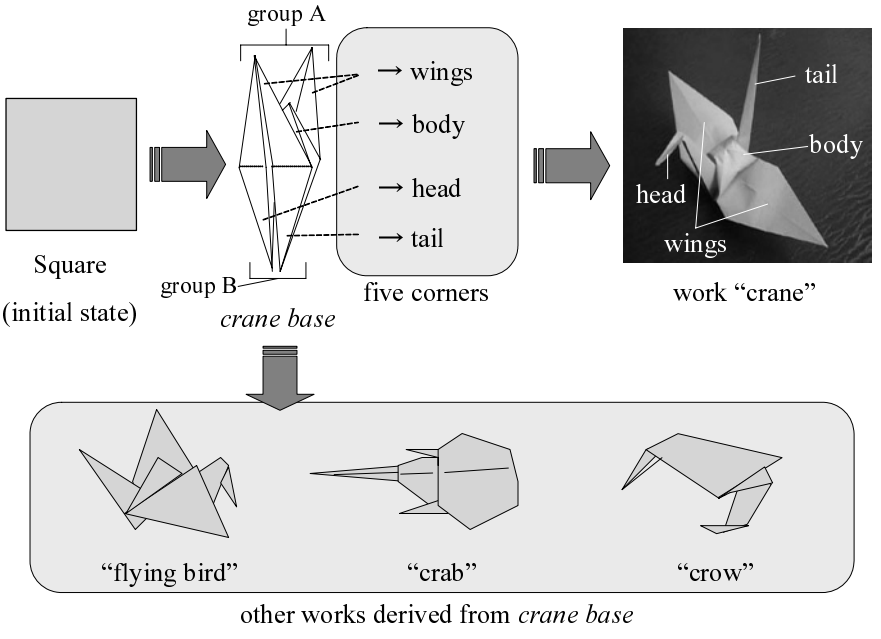


Fig. 7. Example of origami base

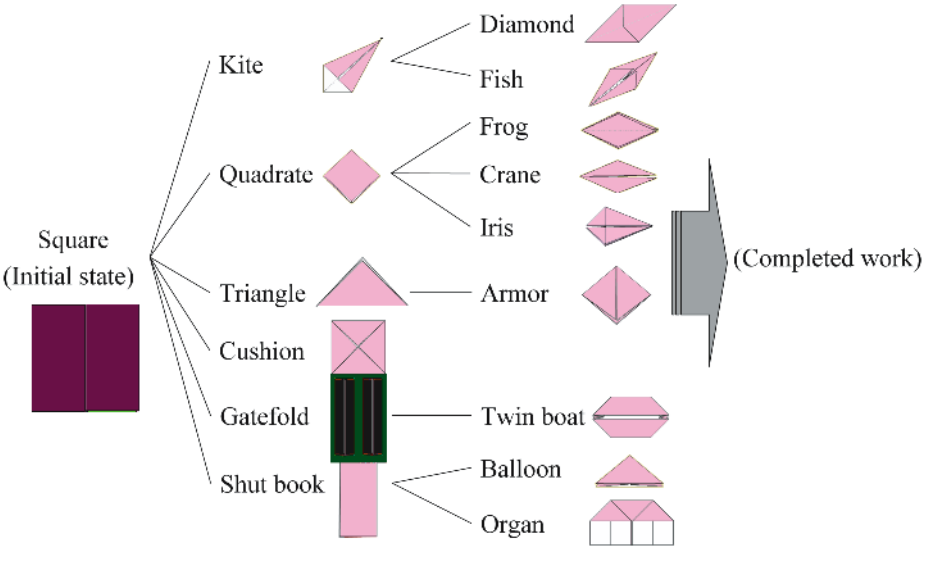


Fig. 8. Derivation graph of origami bases

system should select the origami base corresponding to works of users' intents. Our system teaches the folding process transforming an origami model from square to the origami base. We show how to select the origami base according to users' intents.

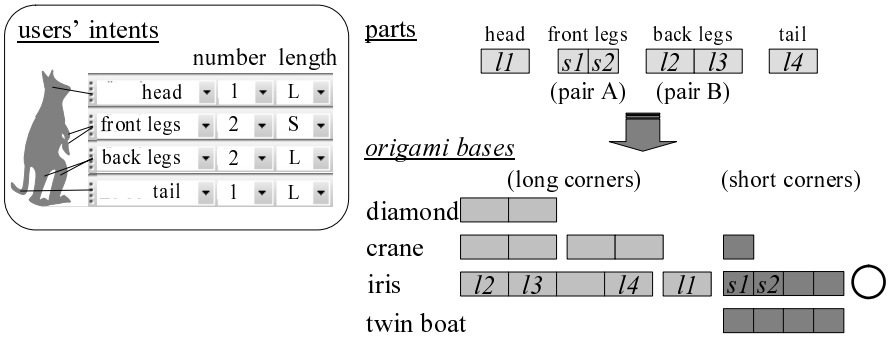


Fig. 9. Example of selecting the origami base

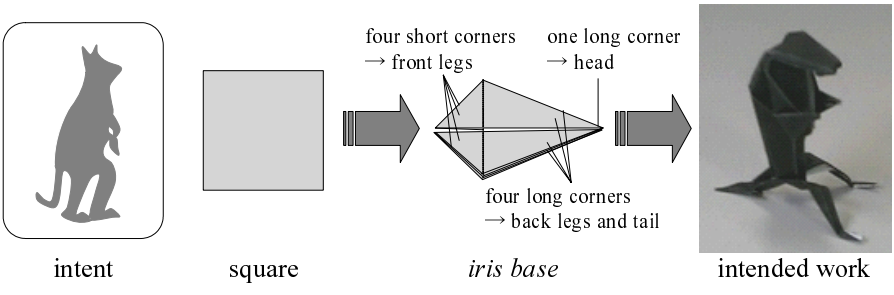


Fig. 10. Example of creating intended work from the base

In users' intended work, there may be a pair of the same kind of parts. The same kind of parts should be derived from the corners of the same group. For example, in the case of Figure 7, wings of the work are a pair. Therefore, it is undesirable that one wing is derived from the corner of group A and the opposite wing is derived from the corner of group B.

From this discuss, the rules for selecting the origami base according to users' intents are as follows.

Rules for Selecting the Origami Base. If a user wants to create a work which has m long parts and n short parts, the system selects the origami bases which satisfy the following conditions.

- Have more than m long corners and more than n short corners.
- Have enough groups which can correspond to each pair of the same kind of parts.

You can consider parts of works as objects and can consider corners of origami bases as containers. In this method, the containers which can accommodate all objects are selected.

Figure 9 shows an example of selecting the origami base according to users' intents. Now, the user wants to create a work of an animal which has two short legs (pair A),

two long legs (pair B), a head, and a short tail. For the sake of simplicity, we assume that there are four origami bases: diamond, crane, iris, and twin boat base. In this case, only iris base is selected, because it has more than four long corners and more than two short corners, and has the group of four long corners corresponding to pair B and the group of four short corners corresponding to pair A. Not having enough corners or groups that can correspond to parts or pairs of the work, other three bases are not selected.

Users can start creating origami works from the selecting origami base which has similar shape to their intended works. Namely, by deriving halfway folding processes, difficulties of users' creation from a square can be overcome. Figure 10 shows an example of creating intended work described above from iris base. The work similar to rough image can be actually created from iris base selected by the system. In this way, it is sure that intended works are easily created from origami base.

5 Conclusions

In this paper, we proposed the system which supports origami creators who have no special knowledge to create their unique works easily in 3-D virtual space. Moreover, the system automatically makes 2-D diagrams or 3-D animation for describing the folding processes so that people can re-build works. Users can decide folding operations and create works by an interactive interface. We discussed three elements about user interface: intended users, cognitive load, and operational error. Consequently, we proposed two methods: a method for representing virtual origami 3-dimensionally, and a method for deriving halfway folding processes by using origami base. By the former method, users can input information about folding operations easily and correctly. By the latter method, users can start creating origami works by themselves. These two methods overcome the difficulties of users' creation of origami works.

As our future work, we must consider advanced methods for deriving halfway folding processes.

Firstly, we should deal with users' complicated intents. For example, when users' intended works have many (more than ten) parts, all existing origami bases can not correspond to them. We consider that this problem is possible to be solved by combination of several origami bases. Actually, there is a basic form called dinosaur base which can be transformed into dinosaurs with lots of parts. Half of this form comes from crane base, and the other half comes from frog base. Namely, a basic form which has more corners may be produced by combining several origami bases. Therefore, we have to consider combination of origami bases.

Secondly, deriving halfway folding process after starting to create must be considered. This paper proposed a method for deriving folding processes from square to some step. However, users may want to vary or add their intents along the way. For this purpose, we consider that the system should recognize where present state are in the derivation graph. Moreover, the learning in the derivation graph will be required.

Finally, we should take into account the characteristics of origami base other than the number, the length, and symmetry of corners. For example, considering alignment of corners according to users' intents, the system will be able to provide more appropriate origami base for users. We must consider what characteristics are useful and how they are input by users.

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